# Evaluation runs: **hermes** (hermes)

generated 2026-11-08 from brain-data b351c947d 2026-11-08 10:25:41 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 7200 | 4652 | conforms | 2.1 | 4 | 4 h, 250 GB | 4 | 7210 | brain01, 2026-11-02 |
| 3,3 | 252865152 | | 🕓 (4 h) | | | 4 h, 250 GB | | | brain01, 2026-11-02 |

## stateful

### stateful: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 3,2 | 7200 | 4652 | conforms | 0.5 | 2 | 4 h, 250 GB | brain01, 2026-11-03 |
| 3,3 | 252865152 | 90010608 | conforms | 2001.3 | 378 | 48 h, 250 GB | brain03, 2026-11-03 |

## offline

### offline: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 7200 | 4652 | conforms | 0.3 | 8 | 4 h, 250 GB | 0.2 | 0.3 | 0.0 | merge | brain01, 2026-11-03 |
| 3,3 | 252865152 | 90010608 | conforms | 1583.8 | 412 | 48 h, 250 GB | 1583.8 | 817.2 | 0.0 | merge | brain03, 2026-11-04 |

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 7200 | 4652 | conforms | 28.6 | 2557 | 4 h, 250 GB | 0.2 | 28.6 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Java implementation against the Rust specification).** skipped: a smaller size had reached its limit: 3 3 bag.

planned but never scheduled (grid points beyond the ladder): 3 3 bag.tfspec-jmcimpl.cinf, 4 2 bag.

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3,2 | 7200 | 4652 | conforms | 20.4 | 2366 | 4 h, 250 GB | | brain01, 2026-11-03 |
| 3,3 | 252865152 | | (4 h) | | | 4 h, 250 GB | | brain01, 2026-11-03 |

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 4652 | 4652 | conforms | 28.6 | 2557 | 4 h, 250 GB | 28.6 | 0.3 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** skipped: a smaller size had reached its limit: 3 3 bag.

planned but never scheduled (grid points beyond the ladder): 3 3 bag.jimpl-tfimpl.cinf, 4 2 bag.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3,2 | 4652 | 4652 | conforms | 28.6 | 2557 | 4 h, 250 GB | 0.3 | 28.6 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: 3 3 bag.

planned but never scheduled (grid points beyond the ladder): 3 3 bag.tfimpl-jimpl.cinf, 4 2 bag.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3,2 | 4652 | 4652 | conforms | 24.4 | 2712 | 4 h, 250 GB | | brain01, 2026-11-03 |
| 3,3 | 90010608 | | (4 h) | | | 4 h, 250 GB | | brain01, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **single**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3,2 | 4652 | 4652 | conforms | 22.0 | 3092 | 4 h, 250 GB | | brain01, 2026-11-03 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).**
skipped: a smaller size had reached its limit: 3 3 bag.